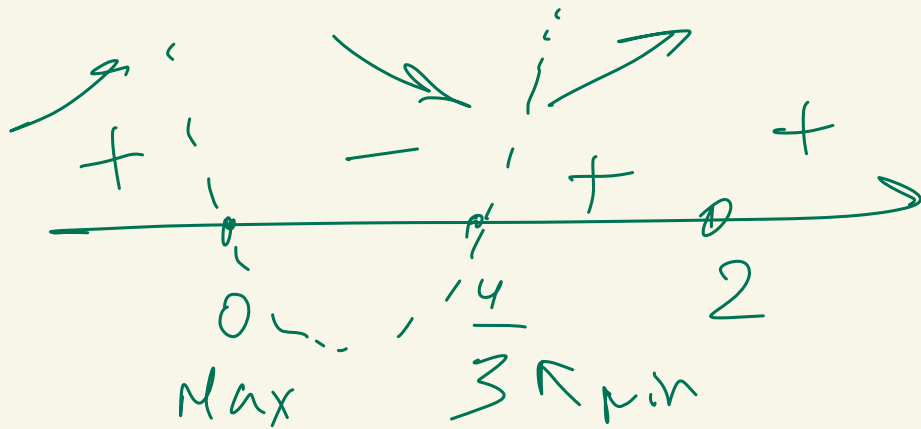






$$g'(x) = \frac{3x^2 - 9x}{(x-2)^2} > 0 \quad x \neq 2$$

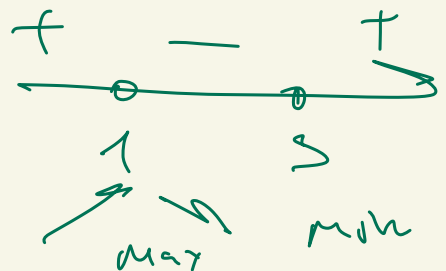
$$y(x) = x^3 - 6x^2 + 9x + 1$$

$$y(1) = 1 - 6 + 9 + 1 \neq 0$$

$$y(-1) = -1 - 6 - 9 + 1 \neq 0$$

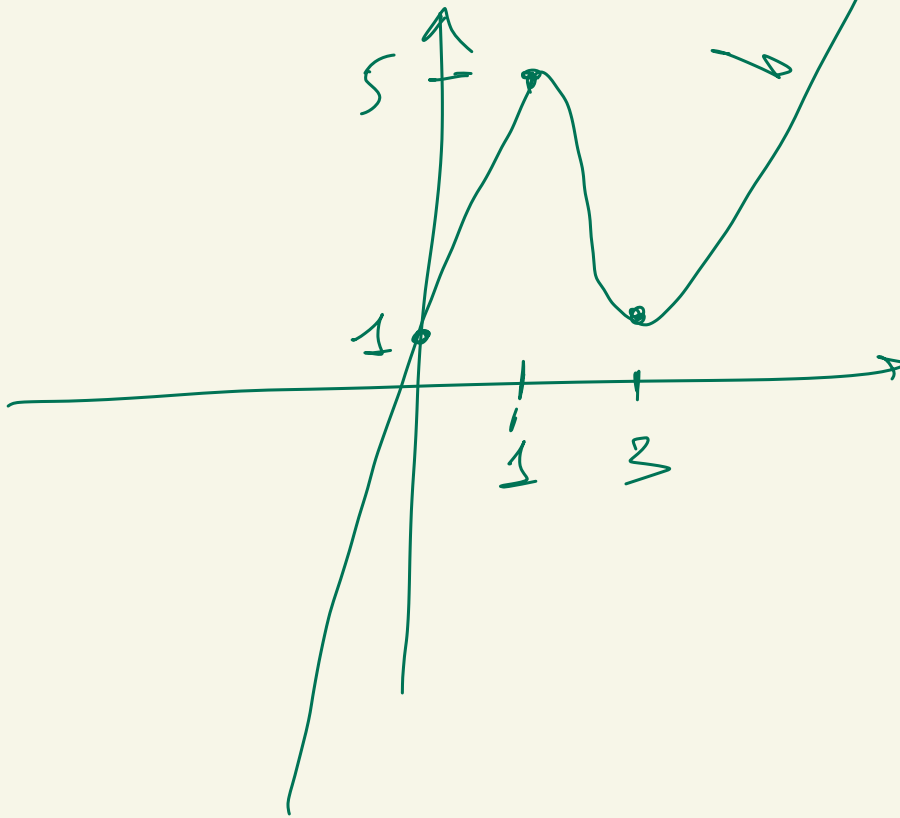
$$y'(x) = 3x^2 - 12x + 9 = 0$$

$$3 : 1$$



$$y(1) = 5$$

$$y(3) = 27 - 54 + 27 + 1 = 1$$



$$y = x^3 - 6x^2 + 11x - 6$$

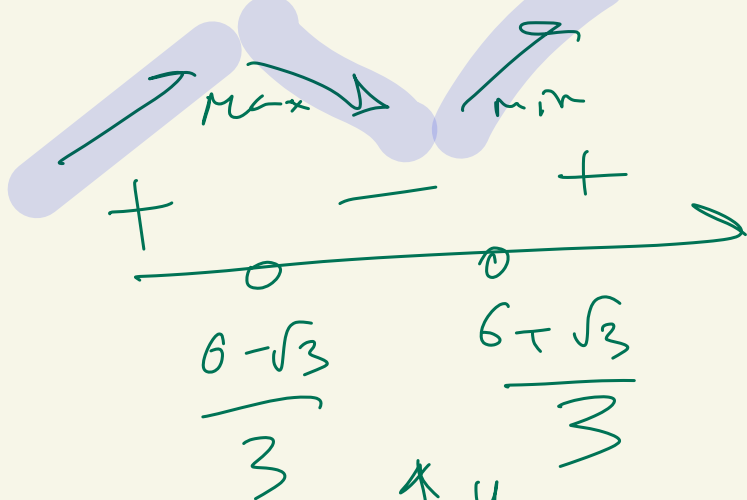
1; 2; 3

$$y' = 3x^2 - 12x + 11$$

$$\frac{12 \pm \sqrt{144 - 3 \cdot 4 \cdot 11}}{6}$$

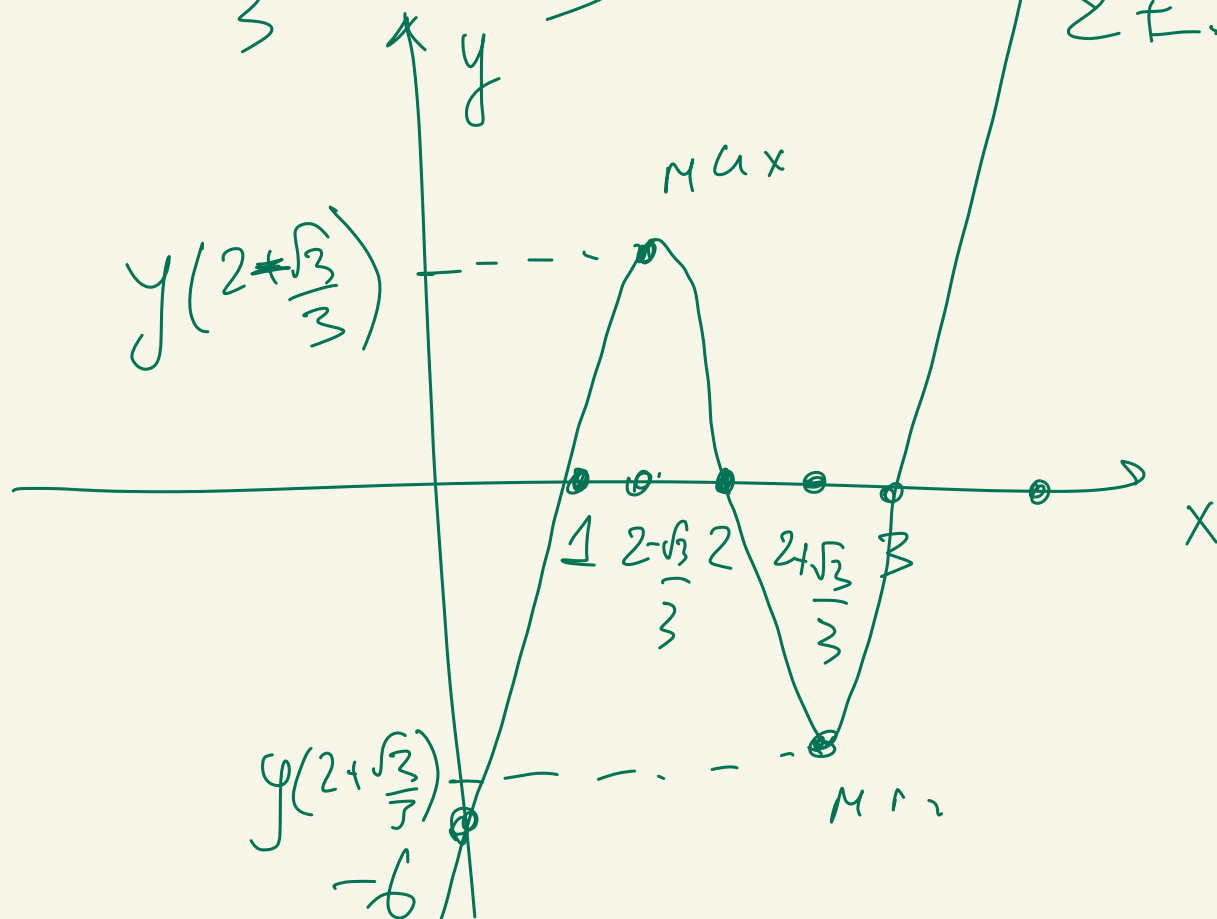
$$\frac{12 \pm \sqrt{144 - 132}}{6}$$

$$\frac{12 \pm \sqrt{12}}{6} =$$



$$\frac{12 \pm 2\sqrt{3}}{3} = \frac{6 \pm \sqrt{3}}{3}$$

$$2 \pm \frac{\sqrt{3}}{3}$$



$$TC = Q^3 - 6Q^2 + 20Q + 50$$

узгелр

$$p = 30$$

$$R(Q) = 3Q$$

$$\pi(Q) = R(Q) - TC(Q)$$

$$TC(Q) ?$$

$$R(Q) ?$$

$$[0; 6]$$

$$SD: \begin{array}{cccc} 1 & 2 & 5 & 10 \\ 1 & -2 & -5 & -10 \end{array}$$

$$1 - 6 + 20 + 50 \quad \cancel{\times}$$

$$-1 - 6 - 20 + 50 \quad \cancel{\times}$$

$$8 - 24 + 40 + 50 \quad \cancel{\times}$$

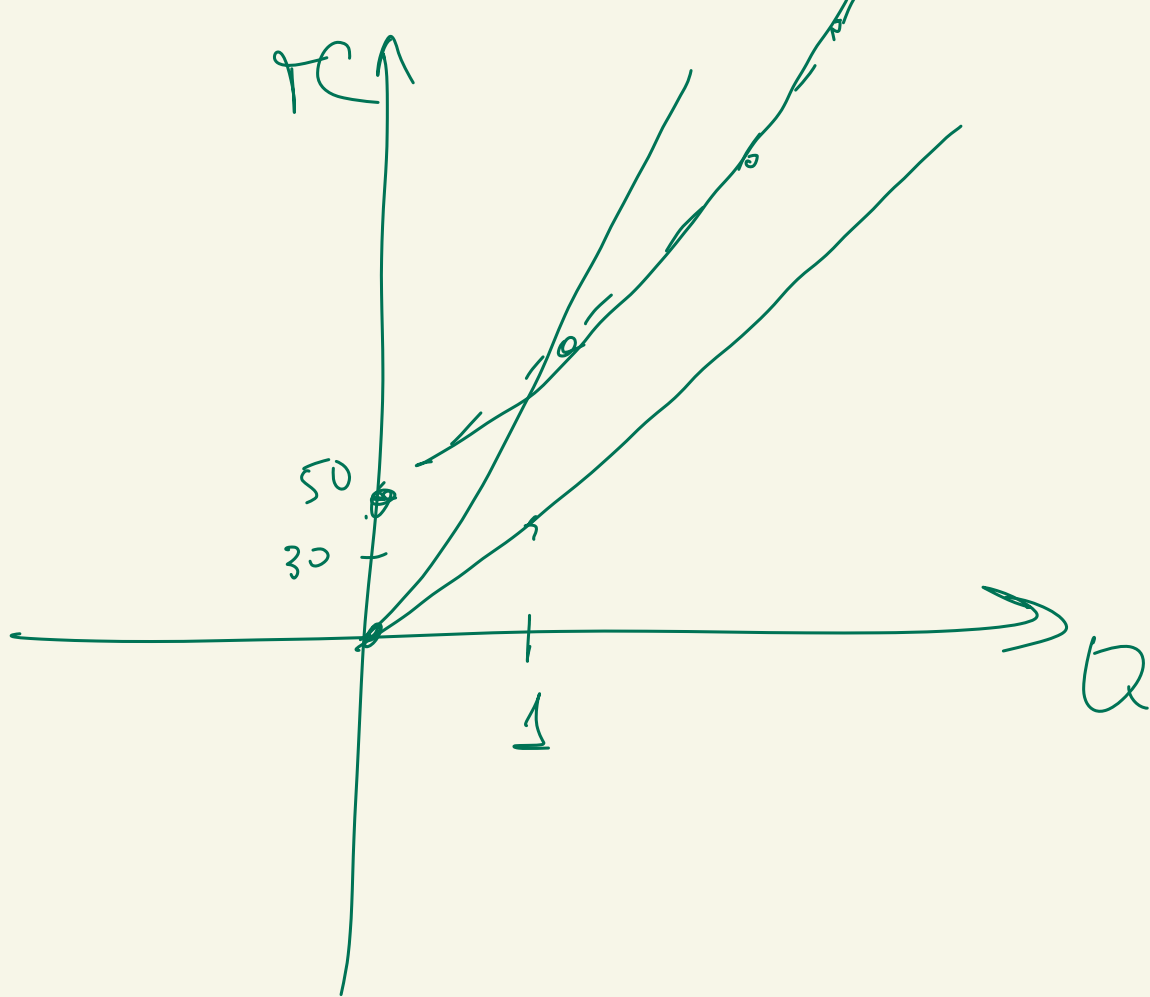
$$-8 - 27 - 40 + 50 \quad -2$$

$$125 - 150 + 100 + 50 \quad \begin{array}{l} 5 \\ -5 \end{array}$$

$$(TC(Q))' = 3Q^2 - 12Q + 20 > 0$$

$$12 \pm \sqrt{12^2 - 80 \cdot 3} < 0$$

/



$$Q^3 - 6Q^2 + 20Q + 50 = 30Q$$

$$Q^3 - 6Q^2 - 10Q + 50$$



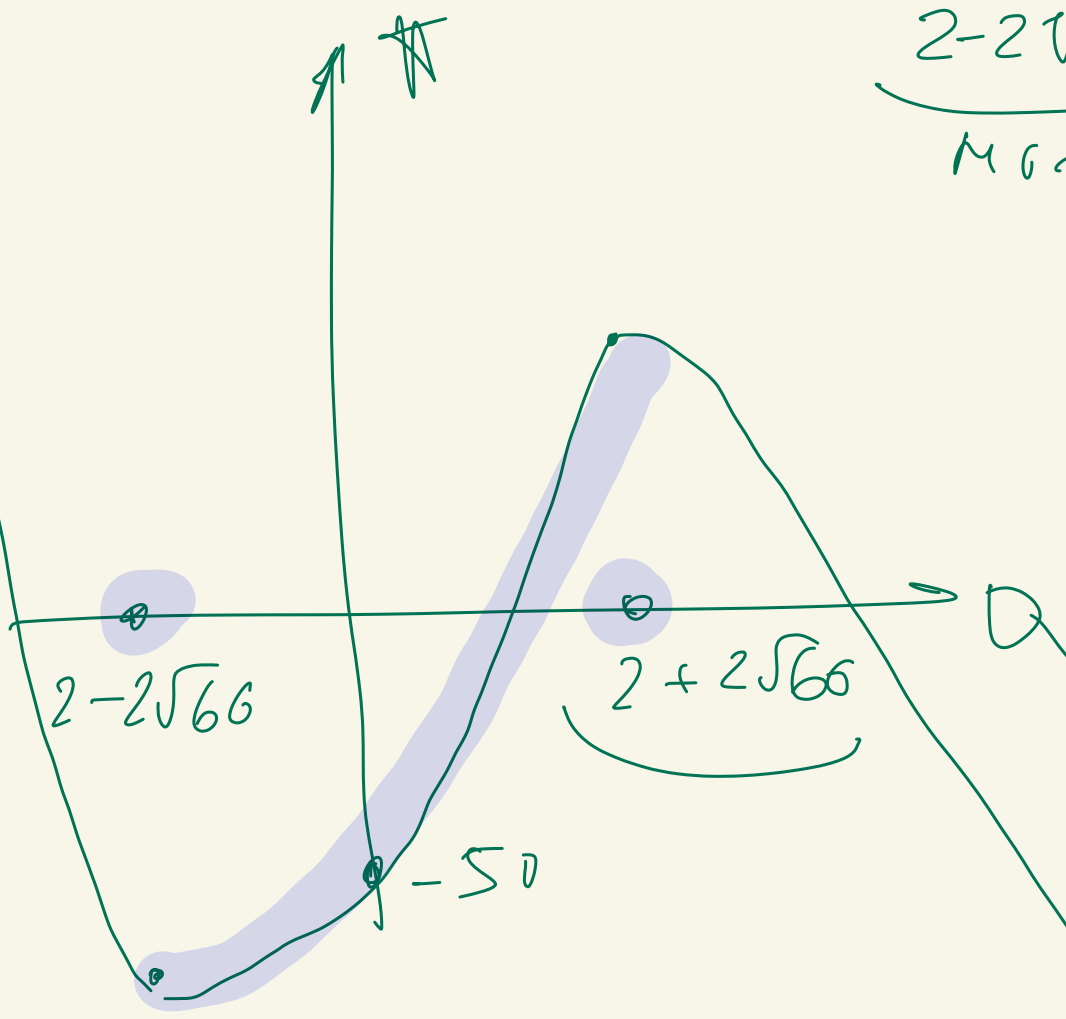
1	2	5
-1	-2	-5

$$\pi(Q) = -Q^3 + 6Q^2 + 10Q - 50$$

$$\pi'(Q) = \frac{-3Q^2 + 12Q + 10 = 0}{-12 \pm \sqrt{144 + 120}}{-6}$$

$$2 \pm \sqrt{264} = 2 \pm 2\sqrt{66}$$

$$\begin{array}{ccc} \swarrow & & \searrow \\ - & + & - \\ \hline 0 & & 0 \\ \underbrace{2 - 2\sqrt{66}}_{\text{Min}} & & \underbrace{2 + 2\sqrt{66}}_{\text{max}} \end{array}$$



no choice 1 to 3 ans.
npy

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{3x}$$

$$= \lim_{x \rightarrow 0} \frac{5x}{3x} =$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{5}{3} = \frac{5}{3}$$

$$x^3 - 8$$

$$\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-1} \right)^x =$$

$$\lim_{x \rightarrow \infty} \left(\frac{x-1+1+2}{x-1} \right)^x =$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x-1} \right)^x = e^3$$

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{(x-2)(x+2)}$$

$$x^3 - 2^3 = (x-2)(x^2 + 2x + 4)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$= \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x + 2} = \frac{4 + 4 + 4}{4}$$

$$= \frac{12}{4} = 3$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{x+3} - 2}{x-1} =$$

$$\lim_{x \rightarrow 1} \frac{(\sqrt{x+3} - 2)(\sqrt{x+3} + 2)}{(x-1)(\sqrt{x+3} + 2)}$$

$$\lim_{x \rightarrow 1} \frac{\cancel{x+3-4}}{\cancel{(x-1)}(\sqrt{x+3} + 2)} =$$

$$\lim_{x \rightarrow 1} \frac{1}{\sqrt{x+3} + 2} = \frac{1}{4}$$

$$1) \lim_{x \rightarrow -2} \frac{x^3 + 8}{x^2 - 4} = \lim_{x \rightarrow -2} \frac{(x+2)(x^2 - 2x + 2)}{(x-2)(x+2)}$$

$$\frac{12}{-8} = -3$$

$$2) \lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4} = \frac{1}{4}$$

$$3) \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$4) \lim_{x \rightarrow 0} \frac{(1+x)^3 - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{3x + 3x^2 + x^3}{x}$$

$$\lim_{x \rightarrow 0} 3 + 3x + x^2 = 3$$

$$\lim_{x \rightarrow 0} \frac{x+x - x-x}{x(\sqrt{1+x} + \sqrt{1-x})} =$$

$$\lim_{x \rightarrow 0} \frac{2x}{x(\sqrt{1+x} + \sqrt{1-x})} =$$

$\lim_{x \rightarrow 0}$

$$\frac{2}{\sqrt{1+x} + \sqrt{1-x}}$$

$=$

$$\textcircled{1}$$